

Assessment of sedation level prior to neonatal intubation: A systematic review

Ellen H. M. de Kort^{1,2}  | Nienke M. Halbmeijer² | Irwin K. M. Reiss² | Sinno H. P. Simons²

¹Department of Pediatrics and Neonatology, Máxima Medical Center, Veldhoven, The Netherlands

²Division of Neonatology, Department of Pediatrics, Erasmus MC Sophia Children's Hospital, Rotterdam, The Netherlands

Correspondence

Ms. Ellen H. M. de Kort, Department of Neonatology, Máxima Medical Center, Veldhoven, The Netherlands.
Email: e.dekort@mmc.nl

Funding information

Fonds NutsOhra; Netherlands Organisation for Health Research and Development, Grant/Award Number: 90713494

Section Editor: Francis Veyckemans

Summary

Background: Adequate premedication before neonatal endotracheal intubation reduces pain, stress, and adverse physiological responses, diminishes duration and number of attempts at intubation, and prevents traumatic airway injury. Therefore, intubation should not be started until an adequate level of sedation is reached. It is not clear how this should be measured in the clinical situation.

Objectives: The aim of this study is to provide a systematic review of the usability and validity of scoring systems or other objective parameters to evaluate the level of sedation before intubation in neonates. Secondary aims were to describe parameters that are used to determine the level of sedation and criteria on which the decision to proceed with intubation is based.

Methods: Literature was searched (January 2017) in the following electronic databases: Embase, Medline, Web of Science, Cochrane Central Registrar of Controlled Trials, Pubmed Publisher, and Google Scholar.

Results: From 1653 hits, 20 studies were finally included in the systematic review. In 7 studies, intubation was started after a predefined time period; in 1 study, pre-oxygenation was the criterion to start with intubation; and in 12 studies, intubation was started in case of adequate sedation and/or relaxation. Only 4 studies described the use of 3 different objective scoring system, all in the neonatal intensive care unit, which are not validated.

Conclusion: No validated scoring systems to assess the level of sedation prior to intubation in newborns are available in the literature. Three objective sedation assessment tools seem promising but need further validation before they can be implemented in research and clinical settings.

KEYWORDS

intubation, newborn infant, premature, premedication, sedation

1 | INTRODUCTION

A significant proportion of preterm and critically ill newborn infants needs endotracheal intubation at some point during their admission to the Neonatal Intensive Care Unit (NICU) or because of anesthesia. Indications range from respiratory insufficiency in case of pulmonary

morbidity, sepsis, or necrotizing enterocolitis, and the administration of surfactant, to surgical procedures such as bowel surgery, surgical closure of a patent ductus arteriosus, laser coagulopathy in case of retinopathy of prematurity, or placement of a surgical central venous catheter. Endotracheal intubation is a painful and stressful procedure and often is associated with adverse physiological responses such as

bradycardia, hypoxemia, systemic hypertension, and intracranial hypertension.¹⁻⁷ The use of premedication before intubation reduces the risk of these adverse events, and also reduces the duration and number of attempts needed for successful intubation and prevents traumatic injury to face, eyes, gums, tongue, and glottic structures.⁶⁻¹¹ Neonatal intubation should therefore be preceded by sedative premedication.

The goal of administering premedication is to achieve a proper level of sedation before intubation is started or muscle relaxants are administered. Still the question is how this level of sedation should be defined and how this should be measured in the clinical situation.

2 | OBJECTIVES

Our primary objective was to provide a systematic review of studies assessing the usability and validity of objective scoring systems or parameters to evaluate the level of sedation before intubation in newborn infants. Because we presumed that very few studies have addressed this issue, our secondary objective was to describe common practice, addressing the following 2 questions:

1. On what parameter(s) and/or criteria is the decision to proceed with intubation after the administration of premedication based?
2. Which (objective) scores or other parameters are used to determine the level of sedation after administration of premedication?

3 | MATERIALS AND METHODS

3.1 | Study design

This systematic review was conducted according to the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).¹²

3.2 | Criteria for considering studies for this review

3.2.1 | Types of studies

For our primary objective, eligible for inclusion were randomized controlled trials (RCTs), quasi-RCTs, controlled clinical trials, and observational studies evaluating the use of objective scoring systems or parameters to assess level of sedation. For our secondary objectives, also single-case studies, poster presentations, editorials, and reviews were eligible for inclusion, provided these extensively reported the complete procedure from administration of premedication to start of the intubation. Availability of full text was imperative.

3.2.2 | Types of study population

The target population consisted of preterm and term neonates who needed endotracheal intubation.

3.2.3 | Types of interventions

For the primary objective, studies evaluating an objective score or parameter to assess the level of sedation after administration of premedication were included. For the secondary objectives, studies addressing any type of intervention were included.

3.2.4 | Types of outcome measures

For the primary objective, studies with degree of sedation during intubation as outcome measure were included. For the secondary objective, studies addressing any type of outcome measure were included.

3.3 | Data collection and analysis

3.3.1 | Electronic searches

The following electronic databases were searched: Embase, Medline, Web of Science, Cochrane Central Registrar of Controlled Trials, Pubmed Publisher, and Google Scholar. The search strategies are outlined in Table 1. There were no restrictions on the basis of publication date or publication status. The search strategy also had no restriction in language. The search was performed on January 27, 2017.

3.3.2 | Selection of studies

Two authors (EdK and NH) independently assessed titles and abstracts of the search results. Full copies of all potentially relevant studies were obtained. Both authors made a decision on final inclusion after retrieval of full copies independently. Disagreements were resolved by consensus; if necessary a third author was consulted (SS). Only articles written in English, Dutch, German, and French were selected.

3.3.3 | Data extraction

Two authors (EdK and NH) extracted details of the included studies independently. The following data were extracted: use of premedication, used drugs or combination of drugs, scores or parameters determining the decision to proceed with intubation, sedation as parameter to proceed with intubation, actions undertaken in case of insufficient sedation.

3.3.4 | Quality appraisal individual studies

Because no studies were found reporting on the usability and validity of scoring systems or other parameters to assess the level of sedation, and we thus only report on common practice as described in the methods sections of the reports, evaluation of the methodological quality and the risk of bias was not relevant.

TABLE 1 Search strategies

Electronic database	Search strategy
Embase	(intubation/de OR "respiratory tract intubation"/exp OR (intubat*):ab,ti) AND (newborn/de OR "newborn intensive care"/exp OR prematurity/de OR "low birth weight"/exp OR "newborn care"/de OR (newborn* OR (new* NEXT/1 born*) OR neonat* OR nicu OR nicus OR prematur* OR preterm* OR "low birth weight" OR lbw OR vlbw OR elbw):ab,ti) AND (premedication/exp OR "hypnotic sedative agent"/exp OR "sedative agent"/exp OR "conscious sedation"/de OR "deep sedation"/de OR "anesthetic agent"/exp OR "muscle relaxation"/de OR "muscle relaxant agent"/exp OR "neuromuscular blocking agent"/exp OR (premedicat* OR sedat* OR anesthetic* OR anaesthetic* OR ((musc* OR neuromusc*) NEAR/3 (relax* OR block*)):ab,ti) NOT ([animals]/lim NOT [humans]/lim) NOT ("cesarean section"/de OR pregnancy/exp OR "pregnant woman"/de OR (cesarean* OR caesarean* OR pregnan*):ab,ti)
Medline	(intubation/OR "Intubation, Intratracheal"/OR (intubat*):ab,ti.) AND (exp infant, newborn/OR "Intensive Care Units, Neonatal"/OR "Intensive Care, Neonatal"/OR (newborn* OR (new* ADJ born*) OR neonat* OR nicu OR nicus OR prematur* OR preterm* OR "low birth weight" OR lbw OR vlbw OR elbw):ab,ti.) AND (premedication/OR exp "Hypnotics and Sedatives"/OR "conscious sedation"/OR "deep sedation"/OR exp "anesthetics"/OR "muscle relaxation"/OR exp "Neuromuscular Agents"/OR (premedicat* OR sedat* OR anesthetic* OR anaesthetic* OR ((musc* OR neuromusc*) ADJ3 (relax* OR block*)):ab,ti.) NOT (exp animals/NOT humans/) NOT ("cesarean section"/OR exp pregnancy/OR "pregnant women"/OR (cesarean* OR caesarean* OR pregnan*):ab,ti.)
Cochrane	((intubat*):ab,ti) AND ((newborn* OR (new* NEXT/1 born*) OR neonat* OR nicu OR nicus OR prematur* OR preterm* OR "low birth weight" OR lbw OR vlbw OR elbw):ab,ti) AND ((premedicat* OR sedat* OR anesthetic* OR anaesthetic* OR ((musc* OR neuromusc*) NEAR/3 (relax* OR block*)):ab,ti) NOT ((cesarean* OR caesarean* OR pregnan*):ab,ti)
Web of Science	TS=(((intubat*))) AND ((newborn* OR (new* NEAR/1 born*) OR neonat* OR nicu OR nicus OR prematur* OR preterm* OR "low birth weight" OR lbw OR vlbw OR elbw)) AND ((premedicat* OR sedat* OR anesthetic* OR anaesthetic* OR ((musc* OR neuromusc*) NEAR/2 (relax* OR block*))) NOT ((cesarean* OR caesarean* OR pregnan*)) NOT ((animal* OR rat OR rats OR mouse OR mice OR murine) NOT (human* OR child* OR patient*)))
Google Scholar	Intubation intubated newborn newborns neonates neonatal nicu nicus premature preterm "low birth weight" lbw vlbw elbw premedication sedation anesthetic anaesthetic "muscular neuromuscular relaxation blockers" -cesarean -caesarean -pregnancy -pregnant

4 | RESULTS

4.1 | Study selection

A flowchart of study selection is provided in Figure 1. The initial electronic database search yielded 2597 records, which number was reduced to 1652 after duplicates were removed. One additional record found in reference lists was added. Thus, the titles and abstracts of in total 1653 records were screened on relevance to the primary and secondary objectives. None of the studies appeared relevant to the primary objective. Regarding the secondary objective, full text was obtained from 75 records and 1578 records were excluded. Of these 75 records, 5 were excluded because of language restriction (1 record) or unavailability (4 records). Of the remaining 70 records, full text was read. The studies' methods sections were screened for information about parameters on which the decision to start with intubation was based. Twenty studies provided this information and these were included in the final analysis of our secondary objective.

4.2 | Study characteristics and results of individual studies

Table 2 shows the characteristics of the 20 included studies. In 12 studies, the decision to proceed with intubation after the administration of premedication was based on the degree of sedation or

relaxation.^{7,13-23} Nevertheless, no more than 4 of these 12 studies used an objective scoring system.¹³⁻¹⁶ In one other of the 12 studies, disappearance of the eyelash reflex was considered an indicator of hypnosis and therefore as the criterion to start with intubation.²¹ In the control group, loss of muscle tone was the criterion to start with intubation.²¹ The methods section did not describe, however, how muscle relaxation was rated. In the remaining 7 studies that based the start of the intubation on the degree of sedation and/or relaxation, it was not clear if sedation and/or relaxation were rated either in an objective or a rather subjective way.^{7,17-20,22,23}

In 8 of the 12 studies, the intubating clinician's observation of insufficient sedation led to the administration of more premedication to achieve an adequate level of sedation and/or relaxation.^{7,13,14,17-19,21,22} In the remaining 4 studies, this was not mentioned at all.^{15,16,20,23}

In 7 of all 20 included studies, intubation was started after a certain period of time had elapsed after the administration of premedication.^{5,6,24-28} The methods section of these studies neither made clear if level of sedation also played a role, nor if all patients were adequately sedated upon start of the intubation procedure. In one study, preoxygenation was the leading parameter to start with intubation.²⁹

Only 4 of the 20 studies used an objective scoring system to assess the level of sedation before proceeding with intubation. These objective scoring systems are outlined in Table 3. The first scoring system is a sedation score used in 2 different reports.

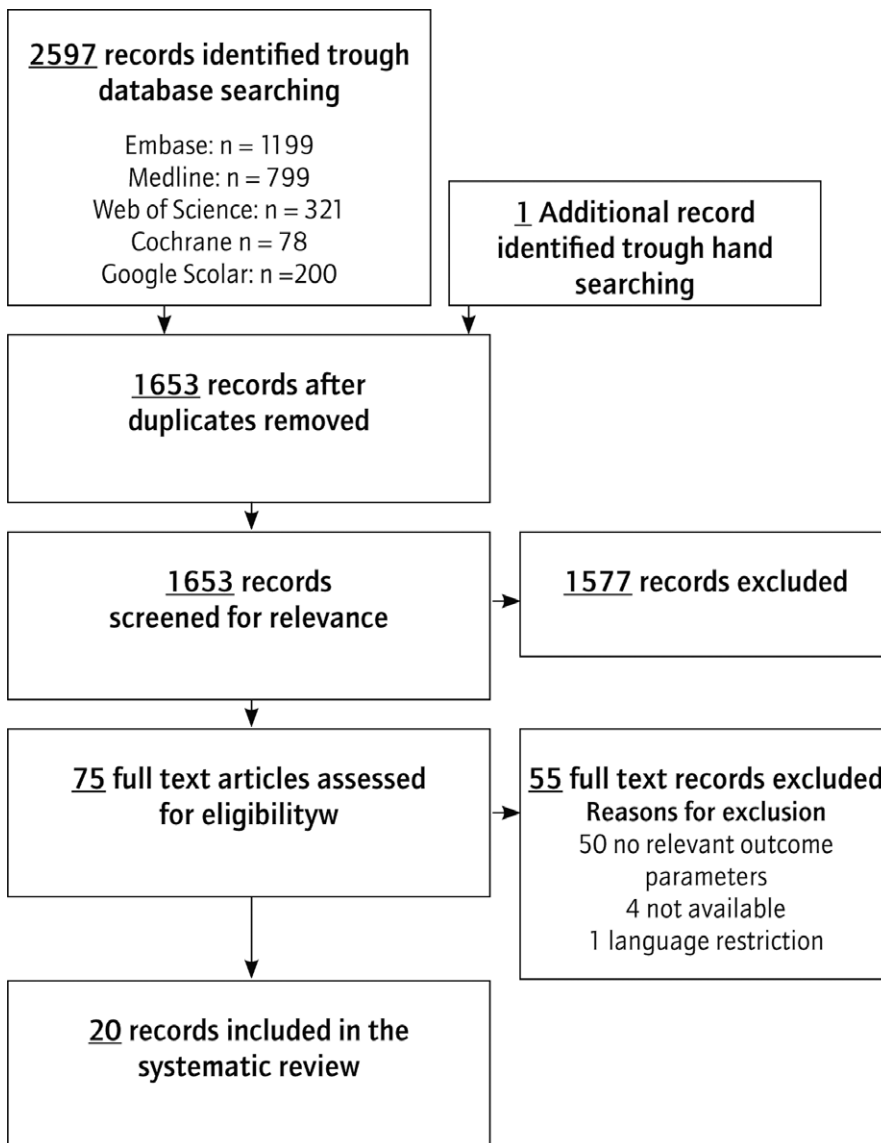


FIGURE 1 Flowchart of included and excluded studies

Smits et al used this score in their dose-finding study for propofol in newborns and de Kort et al used the score in a study on the use of remifentanyl in preterm infants undergoing the INSURE procedure.^{13,14} The score is adopted from the study of Naulaers et al into the effectiveness of methohexital as premedication for elective intubation.¹⁵ The level of sedation is assessed from the motor response to a firm stimulus by rubbing of the feet (1 = spontaneous movement, 2 = moves on touch, 3 = moves on stimulus, 4 = no movement). In both studies, sedation was presumed to be effective with scores 3 and 4, upon which intubation was started.

Another scoring system is the relaxation score also used in the report of Smits et al, combining this score with the above-mentioned sedation score. After the administration of propofol, level of sedation and muscle relaxation were determined. For relaxation, the muscle tone of the extremities was assessed (1 = hypertonic, 2 = normal tone, 3 = mildly hypotonic, and 4 = hypotonic). Also, scores 3 and 4 reflect effective relaxation. Both relaxation and sedation were assessed by the intubating neonatologist. Additional propofol was

administered if sedation and relaxation were found unsatisfactory, until a satisfactory condition was achieved with no predefined maximum dose of propofol.¹³ In the report of de Kort et al,¹⁴ in case of ineffective sedation, additional remifentanyl was administered up to a predefined maximum, with a conversion to propofol in case of persisting inadequate sedation.

The third score is from Thall et al, in their report describing how to perform dose-finding studies for premedication in neonatal intubation. For this purpose they used the Neonatal Pain, Agitation and Sedation Scale (N-PASS) first described by Hummel et al.³⁰ The score consists of 5 variables that should all be scored within 5 minutes of the first sedative administration: crying/irritability, behavior state, facial expression, extremity tone, and vital parameters. Each item is scored on a 5-point scale from −2 to +2, with −2 corresponding to highest sedation and +2 corresponding to highest infant discomfort. For the purpose of endotracheal intubation, Thall et al¹⁶ considered a score between −7 and −3 to reflect a good sedative state.

TABLE 2 Characteristics of included studies

First author, year, country	Study design	Premedication	Parameter for start intubation	Start intubation	Consequences insufficient sedation/relaxation
Smits A, 2016, Belgium	Prospective observational	Propofol	Sedation and relaxation	Good sedation and relaxation scores	Additional propofol in case of unsatisfactory relaxation and sedation according to physician
De Kort, 2016, The Netherlands	Prospective observational	Remifentanyl	Sedation	Good sedation score	Additional remifentanyl until good sedation score
Avino D, 2014, Belgium	Noninferiority randomized trial	SG: remifentanyl CG: morphine, atropine and midazolam	Time	SG: after complete remifentanyl infusion; CG: 3 min after midazolam infusion	
Baleine J, 2014, France	Prospective observational	Nasal midazolam	Sedation	At hypnosis, muscle relaxation or apnea	Second dose in case of arousal at introduction of tube or excessively awake at 5 min
Durrmeyer X, 2014, France	Prospective observational	Atropine, sufentanyl and atracurium	Relaxation	At onset of paralysis	Extra atracurium (up to a predefined maximum)
Thall PF, 2014, USA	Not applicable	Not applicable	Sedation	Good sedation score	Not applicable
Simons S, 2013, The Netherlands	Prospective observational	Propofol	Sedation	When sedation was scored yes by treating physician	Extra propofol (up to a maximum of 3 doses)
Penido MG, 2011, Brazil	Double-blinded randomized trial	SG: propofol, remifentanyl CG: midazolam, remifentanyl	Time	2 min after remifentanyl infusion	
Choong K, 2010, Canada	Double-blinded randomized controlled trial	SG: remifentanyl, atropine and placebo CG: fentanyl, atropine and succinyl-choline	Time	30 s after last drug administration	
Weizling L, 2010, Germany	Prospective observational	Propofol	Time	75 s after propofol administration	
Lemyre B, 2009, Canada	Prospective observational	Atropine, fentanyl and Succinyl-choline (the latter only in GA $\geq$ 34 wks)	Relaxation	Cessation of spontaneous movements (succinyl-choline group) or sufficient relaxation	None mentioned
Van Looy, 2008, USA	Prospective observational	Midazolam and fentanyl	Time	1-2 min after medication administration	Another dose of fentanyl in case of 3 failed attempts
Ghanta S, 2007, Australia	Randomized, open-label controlled trial	SG: propofol CG: morphine, atropine and suxamethonium	Hypnosis and relaxation	SG: disappearance of eyelash reflex CG: muscle relaxation	SG: 1 extra dose of propofol, after that conversion to CG: repeating doses of suxamethonium
Dempsey E, 2006, Ireland	Prospective observational	Mivacurium and fentanyl	Preoxygenation	Saturation > 95%, typically 2 min after fentanyl infusion	
Milesi C, 2006, France	Prospective observational	Nitrous oxide	Relaxation	At suppression of muscle tone (cessation of movements and hypotonia limbs and jaw)	

(Continues)

TABLE 2 (Continued)

First author, year, country	Study design	Premedication	Parameter for start intubation	Start intubation	Consequences insufficient sedation/relaxation
Roberts K, 2006, USA	Randomized controlled trial	SG: atropine, fentanyl, and mivacurium CG: atropine, fentanyl	SG: relaxation CG: time	SG: once spontaneous movements had ceased CG: after fentanyl infusion	None mentioned
Lemyre B, 2004, Canada	Randomized study	SG: morphine CG: placebo	Time	1-2 min after preoxygenation	
Oei J, 2002, Australia	Randomized controlled nonblinded trial	SG: morphine, atropine and suxamethonium CG: none	Relaxation	At disappearance of muscle fasciculations and begin of muscle flaccidity	Up to 4 extra doses of suxamethonium
Buthada A, 2000, USA	Randomized placebo controlled non blinded trial	SG: thiopental CG: placebo	Time	1 min after infusion	
Naulaers G, 1997, Belgium	Prospective observational	Methohexital	Sedation	No reaction to heel rubbing	None mentioned

CG, control group; SG, study group.

5 | DISCUSSION

This systematic review was performed to provide insight into the availability of validated objective scoring systems to assess newborn infants' level of sedation prior to intubation. The literature as per January 27, 2017 does not provide such a validated scoring system.

The decision to proceed with the intubation procedure after administering premedication should be based on the preintubation sedation level. However, studies reporting on this issue show that level of sedation is not always the key factor to proceed with intubation. In several studies intubation proceeded when a certain period of time had elapsed after the administration of premedication, assuming this amount of time being sufficient for drug effect. However, as drug pharmacokinetics and pharmacodynamics can differ considerably depending on gestational age and postnatal age, sufficient drug effect may need a different period of time in individual patients. Using the same time frame in all neonates may very well lead to a proportion of patients not being adequately sedated for the procedure. To guide individualized dosing, we strongly recommend basing the start of the intubation on the actual level of sedation and administering extra doses if necessary.

In the majority of studies included in this review, quality of sedation is indeed the key factor to guide intubation. However, this is assessed in very diverse ways, often rather subjective or not further specified. Validated scoring systems are lacking, especially prior to elective intubation in the operating room. However, the literature describes 3 objective assessment tools used in the NICU: the easy-to-perform sedation score first described by Naulaers et al, the also easy-to-perform relaxation score first used by Smits et al, and the more extensive scoring system suggested by Thall et al which is based on the N-PASS described by Hummel et al.^{13,14,16,30}

The purpose of validated objective scoring systems is to be able to predict effective sedation during the intubation procedure before actually starting the procedure, thereby preventing neonates from intubation without effective sedation. Validation of such scoring systems can be done by comparing the scoring system with the actual level of sedation during the intubation procedure in a larger group of patients. In neonatal intubation studies, the actual level of sedation during an intubation procedure is frequently assessed with the validated intubation score adopted from Viby-Mogensen et al.³¹ In this intubation score, the items ease of laryngoscopy, position of the vocal cords, coughing, jaw relaxation, and movement of the extremities are all judged on a 4-point scale. A score of 2 or less on each item reflects effective sedation.^{13,14,24,25}

Although both using the Viby-Mogenson intubation score to qualify level of sedation during the intubation procedure, the reports of Smits et al and De Kort et al do not provide enough information to draw any conclusions about the accurateness of the sedation and relaxation scores. In the sedation score, the absence of a motor reaction or only a slight motor reaction to a firm stimulus is presumed to indicate the neonate will tolerate the insertion of the endotracheal tube into the supraglottic airway. However, in this assumption, the used stimulus should be stronger than the act of inserting an

TABLE 3 Objective scoring systems

Level of sedation		Highest sedation		Lowest sedation	
Sedation Score	4	3	2	1	
Motor response to firm stimulus	No movements	Movement on firm stimulus	Movement on slight touch	Spontaneous movements	
Definition	Effective sedation = score 3 or 4				
Relaxation score	4	3	2	1	
Extremities tone	Hypotonic	Mildly hypotonic	Normal tone	Hypertonic	
Definition	Effective relaxation = score 3 or 4				
Good sedative state	−2	−1	0	1	2
Crying/irritability	No cry with painful stimuli	Moans or cries minimally with painful stimuli	No sedation/no pain signs	Irritable or crying at intervals, consolable	High-pitched or silent continuous cry, inconsolable
Behavior state	No arousal to any stimuli, no spontaneous movements	Arouses minimally to stimuli, little spontaneous movement	No sedation/no pain signs	Restless, squirming, awakens frequently	Arching, kicking, constantly awake of arouses minimally/no movement
Facial expression	Mouth is lax, no expression	Minimal expression with stimuli	No sedation/no pain signs	Any pain expression intermittent	Any pain expression continual
Extremities tone	No grasp reflex, flaccid tone	Weak grasp reflex, ↓ muscle tone	No sedation/no pain signs	Intermittent clenched toes, fists, finger splay, body is not tense	Continual clenched fists, toes, finger splay, body is tense
Vital signs	No variability with stimuli, hypoventilation or apnea	<1% variability from baseline with stimuli	No sedation/no pain signs	↑ 10%-20% from baseline, SaO ₂ 76%-85% with stimulation—quick recovery	↑ 20% from baseline, SaO ₂ < 75% with stimulation—slow recovery
Definition	Good sedative state = total score on all 5 items −7 to −3				

endotracheal tube. The question rises if the act of inserting an endotracheal tube, mainly via the nasal route, requires a stronger stimulus than heel rubbing. For example, by anesthesiologists, the much firmer stimulus of pinching the trapezius muscle is frequently used before inserting a supraglottic airway. Validation of that stimulus in a clinical study has not been reported. In summary, although both sedation and relaxation scores seem potentially useful scores in the neonatal population, validation of these scores is mandatory before further use.

The scoring system suggested by Thall et al concerns a comprehensive and precise scoring system, which is likely to adequately reflect the level of sedation. Also, different definitions can be used in case of different needs of sedation level according to its purpose. However, to the best of our knowledge, clinical trials using this scoring system are lacking. Therefore, any conclusion about its accuracy and usability is not possible and validation is needed before future use. Even if validated, the scoring system must prove its suitability in daily practice. In neonatal intubation, fast performance of the procedure is mandatory, especially when fast acting agents are used. The extensiveness of the score could possibly make it time consuming and therefore less suitable.

Despite the lack of validated objective scoring systems, several reports do have shown that premedication before neonatal intubation has become standard practice in the majority of neonatal units.^{10,11,32-38} There is however much debate about which premedication or premedication regimen is best. Studies evaluating certain premedication strategies are mainly focused on using 1 dosing strategy for the entire neonatal population. However, pharmacodynamics and pharmacokinetics are influenced by factors such as gestational age, postnatal age and morbidity. For example, Smits et al¹³ found that neonates of different gestational ages needed different doses of propofol for adequate sedation. Most important in our opinion is that the used premedication achieves effective sedation. It should also have a quick recovery to allow for fast extubation, and no significant side effects. The search for the most suitable premedication strategy should be directed toward personalized medicine and focus on administering just enough premedication to achieve adequate sedation in the individual patient. In this point of view, a scoring system that adequately indicates the level of sedation is mandatory.

The work of field and the techniques used differ between pediatric anesthesiologists and neonatologists. For example, they have different drug choices, pursue different levels of anesthesia (deep vs superficial) and require different duration of sedation (hours vs minutes). However, the goals of administering premedication during induction in the operating room and before intubation in the NICU are comparable and in both situations the level of sedation should be objectively assessed before continuing the procedure. With this in mind, neonatologists and anesthesiologists can possibly share valuable knowledge in this area of the field. With the availability of a validated scoring system that can make a distinction between different levels of sedation, it should even be possible to use a single scoring system for both settings.

Limitation of this review is that we used the description of the intubation procedure in the methods section to answer the questions of our secondary objective. Because describing the entire procedure from administration of premedication to intubation was not the primary goal of the included studies, it is possible that these descriptions were not complete and that more scoring systems or other parameters were used in practice.

6 | CONCLUSION

Validated scoring systems to assess the level of sedation prior to intubation in newborns are not available in the literature. Three objective sedation assessment tools seem promising but need further validation before they can be implemented in research and clinical settings. Future research is necessary to find a physical stimulus that enables clinicians to better anticipate the response to insertion of an endotracheal tube and laryngoscopy and to find the best premedication regimen to achieve the preset goals.

ACKNOWLEDGMENTS

We greatly thank Wichor M. Bramer of the Erasmus Medical Center library for performing the electronic database search for this systematic review and Ko Hagoort for editing the manuscript. This research was supported by a grant from Fonds NutsOhra. S. Simons was financially supported by a personal grant of the Netherlands Organisation for Health Research and Development (ZonMw nr 90713494).

DISCLOSURES

The authors report no conflict of interest.

ORCID

Ellen H. M. de Kort  <http://orcid.org/0000-0003-0469-0128>

REFERENCES

1. Maheshwari R, Tracy M, Badawi N, Hinder M. Neonatal endotracheal intubation: how to make it more baby friendly. *J Paediatr Child Health*. 2016;52:480-486.
2. Marshall TA, Deeder R, Pai S, Berkowitz GP, Austin TL. Physiologic changes associated with endotracheal intubation in preterm infants. *Crit Care Med*. 1984;12:501-503.
3. Kelly M, Finer NN. Nasotracheal intubation in the neonate. Physiologic responses and effects of atropine and pancuronium. *J Pediatr*. 1984;105:303-309.
4. Millar C, Bissonnette B. Awake intubation increases intracranial pressure without affecting cerebral blood flow velocity in infants. *Can J Anaesth*. 1994;41:281-287.
5. Lemyre B, Doucette J, Kalyn A, Gray S, Marrin ML. Morphine for elective endotracheal intubation in neonates: a randomized trial. *BMC Pediatr*. 2004;4:20.

6. Buthada A, Sahni R, Rastogi S, Wung JT. Randomised controlled trial of thiopental for intubation in neonates. *Arch Dis Child Fetal Neonatal Ed.* 2000;82:F34-F37.
7. Oei J, Hari R, Butha T, Lui K. Facilitation of neonatal nasotracheal intubation with premedication: a randomized controlled trial. *J Paediatr Child Health.* 2002;38:146-150.
8. Barrington KJ, Finer NN, Etches PC. Succinylcholine and atropine for premedication of the newborn infant before nasotracheal intubation: a randomized, controlled trial. *Crit Care Med.* 1989;17:1293-1296.
9. Carbajal R, Eble B, Anand KJS. Premedication for tracheal intubation in neonates: confusion or controversy? *Semin Perinatol.* 2007;31:30-317.
10. Kumar P, Denson SE, Mancuso TJ, et al. Premedication for nonemergency endotracheal intubation in the neonate. *Pediatrics.* 2010;125:608-615.
11. Byrne E, MacKinnon R. Should premedication be used for semi-urgent or elective intubation in neonates? *Arch Dis Child.* 2006;91:79-83.
12. Moher D, Lliberati A, Tetzlaff J, Altman GG; PRISMA Group. Preferred reporting items for systematic reviews and meta-analysis: the PRISMA statement. *J Clin Epidemiol.* 2009;62:1006-1012.
13. Smits A, Thewissen L, Caicedo A, Naulaers G, Allegaert K. Propofol dose-finding to reach optimal effect for (semi-)elective intubation in neonates. *J Pediatr.* 2016;179:54-60.
14. De Kort EHM, Hanff L, Roofthoof D, Reiss IKM, Simons SHP. Insufficient sedation and severe side effects after fast administration of remifentanyl during INSURE in preterm newborns. *Neonatology.* 2017;111:172-176.
15. Naulaers G, Deloof E, Vanhole C, Kola E, Devlieger H. Use of methohexital for elective intubation in neonates. *Arch Dis Child.* 1997;77:F61-F64.
16. Thall PF, Nguyen HQ, Zohar S, Maton P. Optimizing sedative dose in preterm infants undergoing treatment for respiratory distress syndrome. *J Am Stat Assoc.* 2014;109:931-943.
17. Baleine J, Milesi C, Mesnage R, et al. Intubation in the delivery room: experience with nasal midazolam. *Early Hum Dev.* 2014;90:39-43.
18. Durrmeyer X, Dahan S, Delorme P, et al. Assessment of atropine-sufentanil-atracurium anaesthesia for endotracheal intubation: an observational study in very premature infants. *BMC Pediatr.* 2014;14:120.
19. Simons SHP, Van der Lee R, Reiss IKM, van Weissenbruch MM. Clinical evaluation of propofol as sedative for endotracheal intubation in neonates. *Acta Paediatr.* 2013;1002:e487-e492.
20. Lemyre B, Cheng R, Gaboury I. Atropine, fentanyl and succinylcholine for non-urgent intubations in newborns. *Arch Dis Child Fetal Neonatal Ed.* 2009;94:F436-F442.
21. Ghanta S, Abdel-Latif ME, Lui K, Ravindranathan H, Awas J, Oie J. Propofol compared with the morphine, atropine and suxamethonium regimen as induction agents for neonatal endotracheal intubation: a randomized, controlled trial. *Pediatrics.* 2007;119:1248-1255.
22. Milesi C, Pidoux O, Sabatier E, Badr M, Cambonie G, Picaud JC. Nitrous oxide analgesia for intubating preterm neonates: a pilot study. *Acta Paediatr.* 2006;95:1104-1108.
23. Roberts KD, Leone TA, Edwards WH, Rich WD, Finer NN. Premedication for nonemergent neonatal intubations: a randomized, controlled trial comparing atropine and fentanyl to atropine, fentanyl, and mivacurium. *Pediatrics.* 2006;118:1583-1591.
24. Avino D, Zhang WH, De Villé A, Johansson AB. Remifentanyl versus morphine-midazolam premedication on the quality of endotracheal intubation in neonates: a noninferiority randomized trial. *J Pediatr.* 2014;164:1032-1037.
25. Penido MG, De Oliveira Silva DF, Tavares EC, Silva YP. Propofol versus midazolam for intubating preterm neonates: a randomized controlled trial. *J Perinatol.* 2011;31:356-360.
26. Choong K, Alfaleh K, Doucette J, et al. Remifentanyl for endotracheal intubation in neonates: a randomized controlled trial. *Arch Dis Child Fetal Neonatal Ed.* 2010;95:F80-F84.
27. Welzing L, Kribs A, Eifinger F, Huenseler C, Oberthuer A, Roth B. Propofol as an induction agent for endotracheal intubation can cause significant arterial hypotension in preterm neonates. *Pediatr Anesth.* 2010;20:605-611.
28. Van Looy JW, Schumacher RE, Bhatt-Mehta V. Efficacy of a premedication algorithm for nonemergent intubation in a neonatal intensive care unit. *Ann Pharmacother.* 2008;43:947-955.
29. Dempsey EM, Hazzani FA, Faucher D, Barrington KJ. Facilitation of neonatal endotracheal intubation with mivacurium and fentanyl in the neonatal intensive care unit. *Arch Dis Child Fetal Neonatal Ed.* 2006;91:F279-F282.
30. Hummel P, Puchalski M, Creech SD, Weis MG. Clinical reliability and validity of the N-PASS: neonatal pain, agitation and sedation scale with prolonged pain. *J Perinatol.* 2008;28:55-60.
31. Viby-Mogensen J, Engbaek J, Eriksson LI, et al. Good clinical research practice (GCRP) in pharmacodynamics studies of neuromuscular blocking agents. *Acta Anaesthesiol Scand.* 1996;40:59-74.
32. Whyte S, Birrell G, Wyllie J. Premedication before intubation in UK neonatal units. *Arch Dis Child Fetal Neonatal Ed.* 2000;82:F38-F41.
33. Kelleher J, Mallya P, Wyllie J. Premedication before intubation in UK neonatal units: a decade of change? *Arch Dis Child Fetal Neonatal Ed.* 2009;94:F332-F335.
34. Chaudhary R, Chonat S, Gowda H, Clarke P, Curley A. Use of premedication for intubation in tertiary neonatal units in the United Kingdom. *Pediatr Anesth.* 2009;19:653-658.
35. Wheeler B, Broadbent R, Reith D. Premedication for neonatal intubation in Australia and New Zealand: a survey of current practice. *J Paediatr Child Health.* 2012;48:997-1000.
36. Durrmeyer X, Daoud P, Decobert F, et al. Premedication for neonatal endotracheal intubation: results from the epidemiology of procedural pain in neonates study. *Pediatr Crit Care Med.* 2013;14:e169-e175.
37. Simon L, Trifa M, Mokhtari M, Hamza J, Treluyer JM. Premedication for tracheal intubation: a prospective survey in 75 neonatal and pediatric intensive care units. *Crit Care Med.* 2004;32:565-568.
38. Sarkar S, Schumacher RE, Baumgart S, Donn SM. Are newborns receiving premedication before elective intubation. *J Perinatol.* 2006;26:286-289.

How to cite this article: de Kort EHM, Halbmeijer NM, Reiss IKM, Simons SHP. Assessment of sedation level prior to neonatal intubation: A systematic review. *Pediatr Anesth.* 2017;00:1-9. <https://doi.org/10.1111/pan.13285>